

Types in concurrency



Emilio Tuosto @ GSSI

Reliable Software by Construction

5-10 October
Lorentz Center, Leiden

– My assignment –

[...]

Tutorials are intended to be 90 minutes long. The goal is to create sufficient common knowledge that will enable interactive discussions later on.

We have in mind a broad exposition of a specific development on the topic, ideally with pointers to the relevant literature.

From Jorge's email

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[...]

audience includes people from academia, but also practitioners from industry
who may not be familiar with the style/notation typical of formal methods presentations.

A non-exhaustive list

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EDITORIAL

Rocco De Nicola · Davide Sangiorgi

Types in concurrency

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Traditionally types have played a very limited role in concurrency; they were essentially used for specifying the nature of the exchanged values. The picture has changed after the introduction of formalisms for dealing with systems of mobile processes. In addition to the classical use of types in sequential languages, e.g. for static detection of run time error, enhancing program readability, memory management, abstracting from implementation details, types have emerged as an important tool for:

- specifying interfaces and possible interactions
- controlling process mobility
- controlling resource usage
- controlling access rights
- developing new proof techniques
- improving efficiency of verification algorithms
- guaranteeing integrity and confidentiality of data

"(4) to make programs easier to understand" [24, p. 233]

...and more recently, e.g.,

- ▶ foundations
 - ▶ connection with logics
 - ▶ expressivity
 - ▶ new models of coordination
- ▶ programming
- ▶ static analysis
- ▶ run-time verification (monitoring)
- ▶ information flow analysis
- ▶ subtyping
- ▶ ...

A taste of program comprehension

Erlang

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Let's get some help...

Friendlier representations with (formal) models

Local behaviour [18, 20, 4]



CFSMs: FIFO buffers as well

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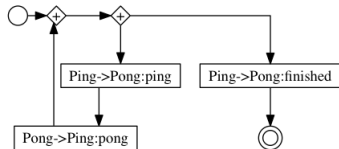


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Global behaviour [20, 11, 23]

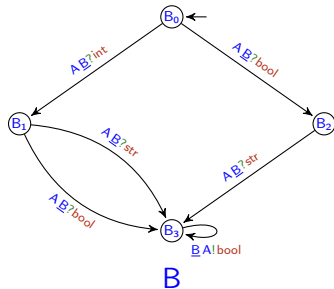
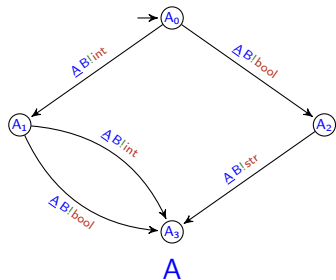


... "synchronous" distributed workflow

$$\mu t. C \rightarrow S : \begin{cases} \text{finished} \\ \text{ping}.S \rightarrow C : \text{pong}.t \end{cases}$$

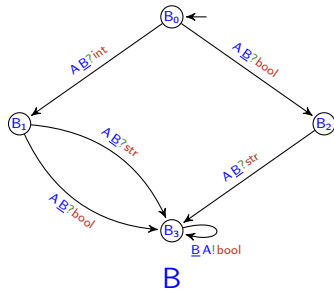
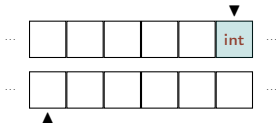
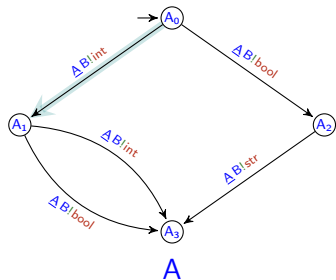
A formal model...informally

Communicating systems



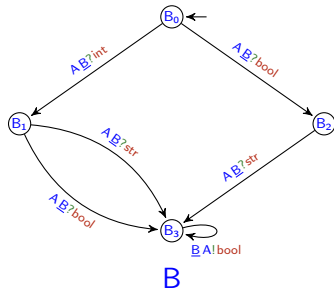
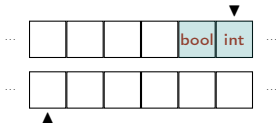
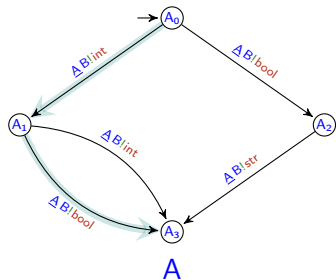
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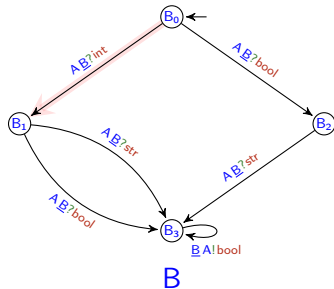
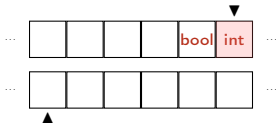
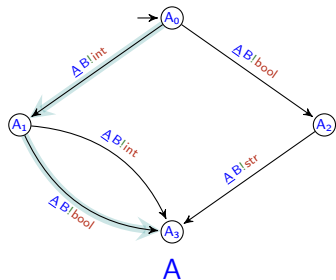
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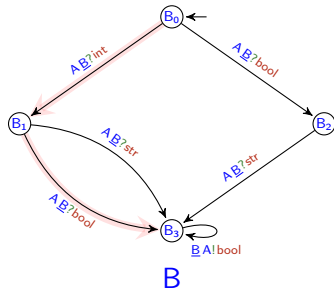
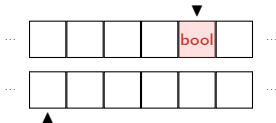
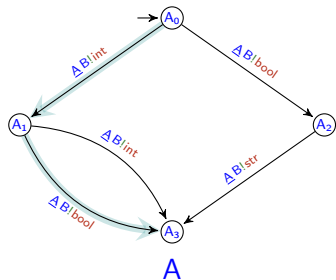
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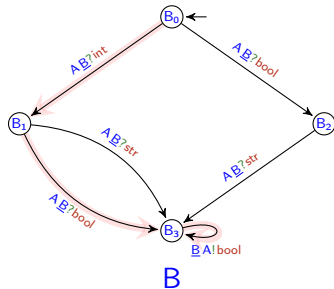
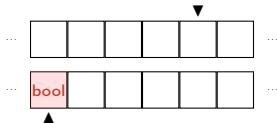
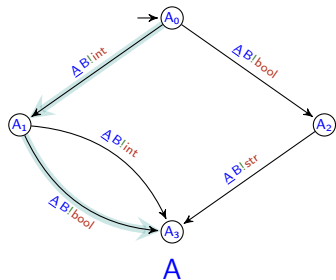
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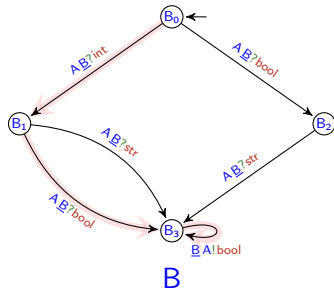
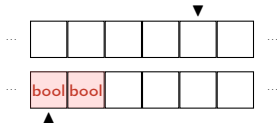
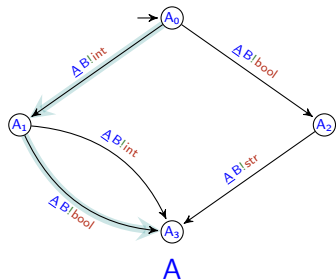
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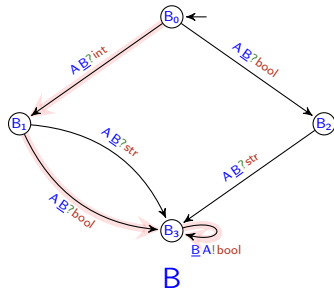
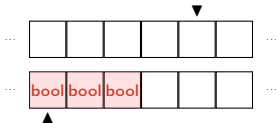
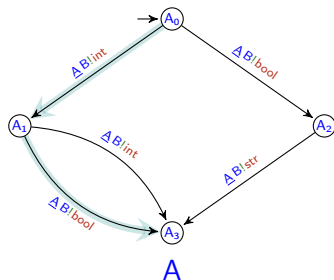
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- ▶ a **communicating finite-state machine** (CFSM) is an FSA^a whose transitions are input/output actions executed by a single participant and whose states are all accepting
- ▶ a **communicating system** is a finite map^b assigning to a participant **A** a CFSM executing communications of **A**

^aAkin (UML) state machine

^bA.k.a. dictionary

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So what?

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Is the program terminating?

Concurrency/distribution is challenging!

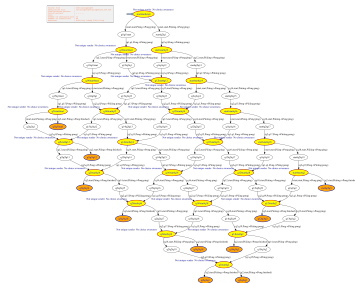
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– Choreographies –

"Top-down"

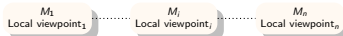
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Synchrony

Choreography G
global viewpoint

Asynchrony



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M_1
Local viewpoint₁

M_i
Local viewpoint_i

M_n
Local viewpoint_n

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Well-formedness

Asynchrony

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Local viewpoint₁

M_j
Local viewpoint_j

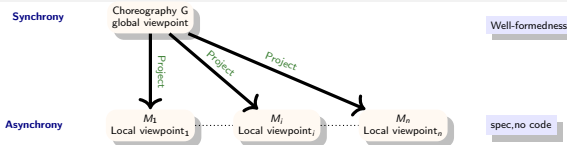
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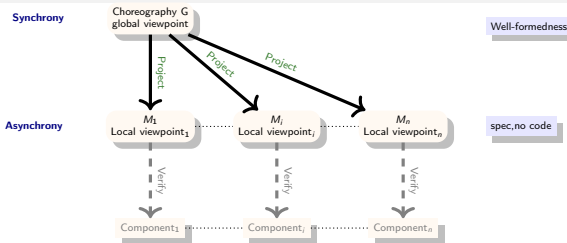
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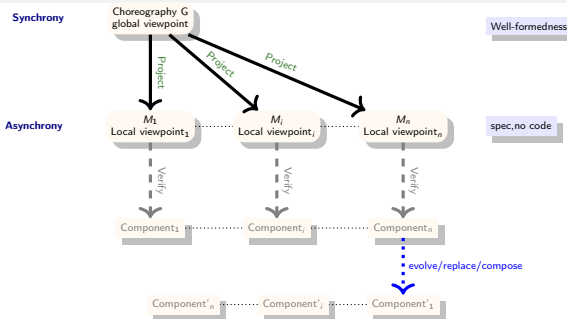
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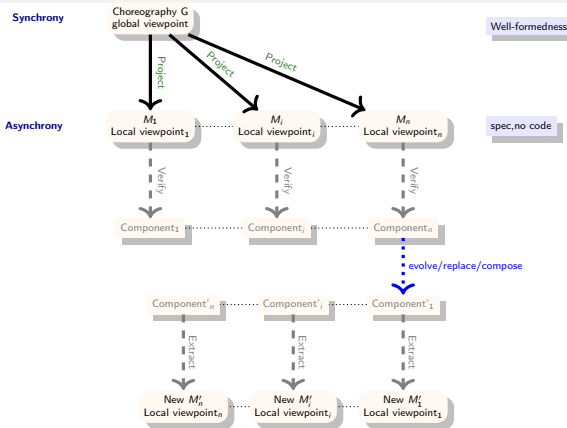
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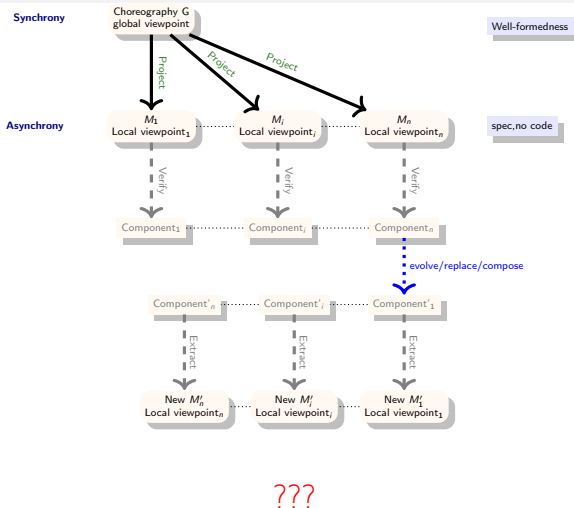
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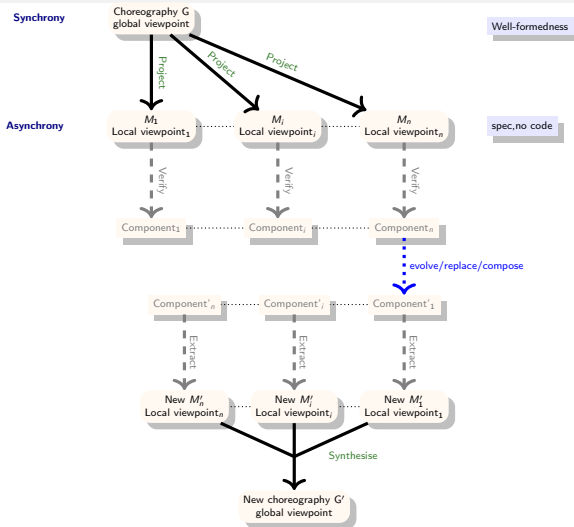
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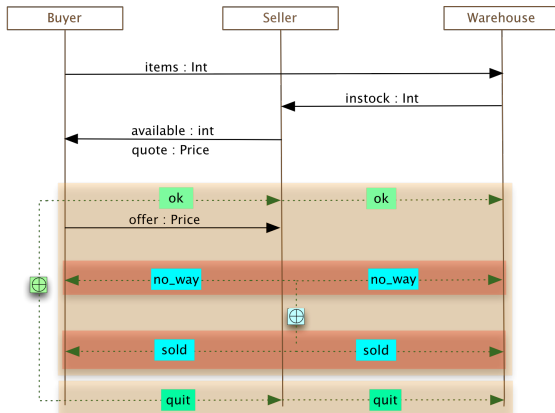
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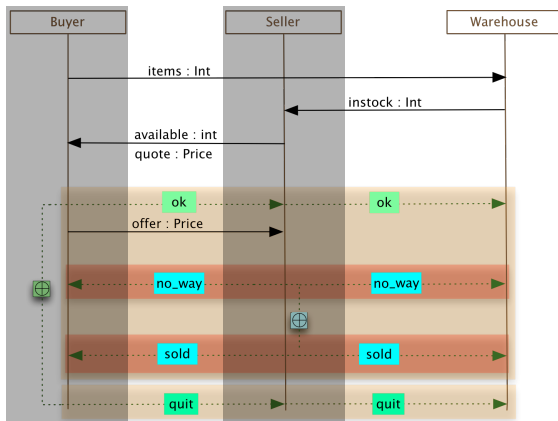


An intuitive account...



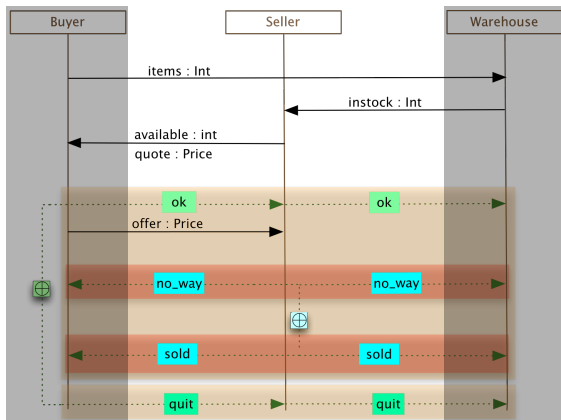
Global viewpoint

An intuitive account...



Projecting on Warehouse

An intuitive account...



Projecting on **seller**

– Design-by-contract of protocols –

A formal global view

global types [20]:

$$\begin{array}{lcl} \mathcal{G} & ::= & A \rightarrow B : m \\ & | & A \rightarrow B : \left(l_j : \mathcal{G}_j \right)_{j \in J} \\ & | & \mu t . \mathcal{G} \\ & | & t \langle \tilde{e} \rangle \\ & | & \text{end} \end{array}$$

A formal global view

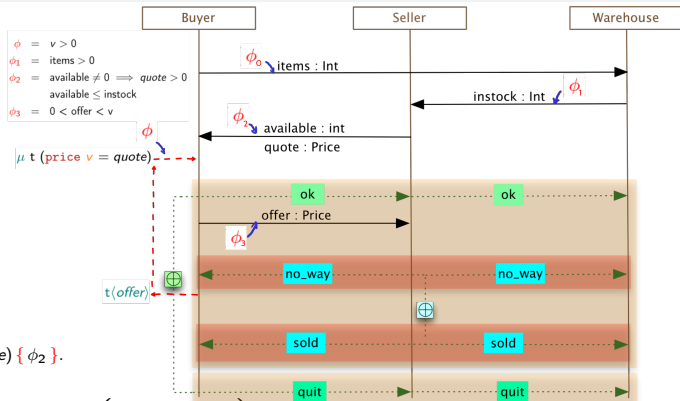
Asserted [2] global types [20]:

$$\begin{array}{lcl} \mathcal{G} & ::= & A \rightarrow B: (\tilde{s} \tilde{x}) \{ \phi \}. \mathcal{G} \\ & | & A \rightarrow B: \left(\{ \phi_j \} l_j: \mathcal{G}_j \right)_{j \in J} \\ & | & \mu t (\tilde{s} \tilde{x} = \tilde{e}) \{ \phi \}. \mathcal{G} \\ & | & t \langle \tilde{e} \rangle \\ & | & \text{end} \end{array}$$

where

- ▶ ϕ and ϕ_j are predicates constraining variables x occurring in the protocol
- ▶ l_j are labels
- ▶ e is an expression denoting values of basic sorts s (eg integer, lists, bool, etc)
- ▶ recursion parameters are initialised with some values

DbC on our buyer-seller-warehouse protocol



$\mathcal{G}_{neg} =$ Buyer $\rightarrow$ Warehouse: (int items) $\{\phi_0\}$.
 Warehouse $\rightarrow$ Seller: (int instock) $\{\phi_1\}$.
 Seller $\rightarrow$ Buyer: (int available, price quote) $\{\phi_2\}$.
 μt (price v = quote) $\{\phi\}$.

Buyer $\rightarrow$ Seller: $\left(\begin{array}{l} \{T\}ok: \text{Buyer} \rightarrow \text{Warehouse}: \left(\begin{array}{l} \{T\}ok: \mathcal{G}_{ok} \end{array} \right), \\ \{T\}quit: \text{Buyer} \rightarrow \text{Warehouse}: \left(\begin{array}{l} \{T\}quit: \text{end} \end{array} \right) \end{array} \right)$

$\mathcal{G}_{ok} =$ Buyer $\rightarrow$ Seller: (price offer) $\{0 < \text{offer} < v\}$.

Seller $\rightarrow$ Buyer: $\left(\begin{array}{l} \{T\}no_way: \text{Seller} \rightarrow \text{Warehouse}: \left(\begin{array}{l} \{T\}no_way: t\langle offer \rangle \end{array} \right), \\ \{T\}sold: \text{Seller} \rightarrow \text{Warehouse}: \left(\begin{array}{l} \{T\}sold: \text{end} \end{array} \right) \end{array} \right)$

Bad specs

Exercise

What's wrong with the following specifications?

$$\mathcal{G}_1 = A \rightarrow B: (\text{int } x) \{ \top \}. B \rightarrow C: (\text{int } y) \{ \top \}. C \rightarrow A: (\text{int } v) \{ v > x \}. \text{end}$$

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History Sensitivity In any interaction of $A \rightarrow B: (s \ x) \{ \phi \}. \mathcal{G}$ in G A "knows" all the variables in ϕ

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$\mathcal{G}_2 = A \rightarrow B: (\text{int } x) \{ x \geq 6 \}. B \rightarrow A: (\text{int } y) \{ y \leq x \wedge y > 6 \}. \text{end}$

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Temporal Satisfiability Each predicate ϕ in G is satisfiable under the constraints "accumulated" by the predicates of interactions/branchings occurring before A in G

Endpoints & projections

$$\begin{aligned}
 \mathcal{T} \quad ::= & \quad !R(\mathbf{s} \ \mathbf{x})\{\phi\};\mathcal{T} \mid \oplus R \left(\{\phi_i\}l_i : \mathcal{T}_i \right)_{i \in I} \\
 & \mid ?S(\mathbf{s} \ \mathbf{x})\{\phi\};\mathcal{T} \mid \&S \left(\{\phi_i\}l_i : \mathcal{T}_i \right)_{i \in I} \\
 & \mid \mu \mathbf{t} \ (\mathbf{s} \ \mathbf{x} = \mathbf{e})\{\phi\}. \mathcal{T} \\
 & \mid \mathbf{t}\langle \mathbf{e} \rangle \\
 & \mid \text{end}
 \end{aligned}$$

$$S \rightarrow R: (\mathbf{s} \ \mathbf{x})\{\phi'\}.\mathcal{G}' \downarrow_A^\phi = \begin{cases} !R(\mathbf{s} \ \mathbf{x})\{\phi'\}; (\mathcal{G}' \downarrow_A^{\phi \wedge \phi'}) & \text{if } A = S \\ ?S(\mathbf{s} \ \mathbf{x})\{(\phi \wedge \phi') \downarrow_A\}; (\mathcal{G}' \downarrow_A^{\phi \wedge \phi'}) & \text{if } A = R \\ \mathcal{G}' \downarrow_A^{\phi \wedge \phi'} & \text{otherwise} \end{cases}$$

$$S \rightarrow R: \left(\{\phi_i\}l_i : \mathcal{G}_i \right)_{i \in I} \downarrow_A^\phi = \begin{cases} \oplus R \left(\{\phi_i\}l_i : \mathcal{G}_i \downarrow_A^{\phi \wedge \phi_i} \right)_{i \in I} & \text{if } A = S \\ \&S \left(\{(\phi \wedge \phi_i) \downarrow_A\}l_i : \mathcal{G}_i \downarrow_A^{\phi \wedge \phi_i} \right)_{i \in I} & \text{if } A = R \\ \sqcup_{i \in I} \mathcal{G}_i \downarrow_A^{\phi \wedge \phi_i} & \text{otherwise} \end{cases}$$

...

A glimpse of the typing

A glimpse of the typing: initiate

$$\frac{\Gamma \vdash a : \mathcal{G} \quad \mathcal{C} ; \Gamma \vdash P \triangleright \Delta, c : \mathcal{G} \downarrow_A}{\mathcal{C} ; \Gamma \vdash A @ \bar{a}(c). P \triangleright \Delta}$$

A glimpse of the typing: initiate, join

$$\frac{\Gamma \vdash a : \mathcal{G} \quad \mathcal{C} ; \Gamma \vdash P \triangleright \Delta, c : \mathcal{G} \downarrow_A}{\mathcal{C} ; \Gamma \vdash A @ \bar{a}(c).P \triangleright \Delta}$$

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A glimpse of the typing: initiate, join, start

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$$\frac{\Gamma \vdash e : \mathbf{s} \quad \mathcal{C} \supset \phi[e/x] \text{ valid} \quad \mathcal{C} ; \Gamma \vdash P[e/x] \triangleright \Delta, c : \mathcal{T}[e/x]}{\mathcal{C} ; \Gamma \vdash X @ c ! e ; P \triangleright \Delta, c : ! X(\mathbf{s} \ x) \{ \phi \} ; \mathcal{T}}$$

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.....

Thm. Well-typed processes behave according to the protocols they implement

– Semantics of choreographies –

Making sense of global views

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- ▶ The previous theorem (and the others in [2]) leverage projections & an operational semantics of local types

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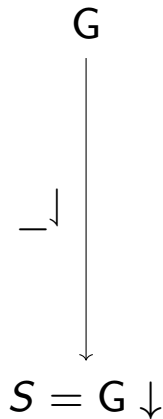
- ▶ The previous theorem (and the others in [2]) leverage projections & an operational semantics of local types
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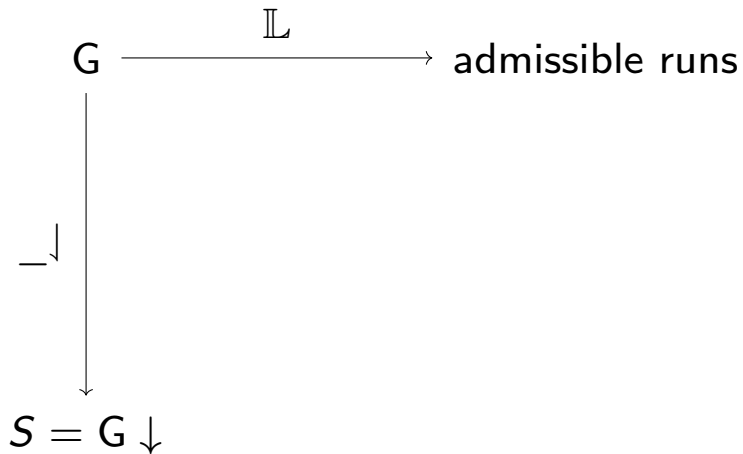
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- ▶ Other approaches introduce an operational semantics of global types (see e.g., [5, 22, 20, 9, 10, 3])
- ▶ Recently there has been interest on denotational semantics of global views (see e.g., [25, 16, 7, 14, 12, 8, 13])

G

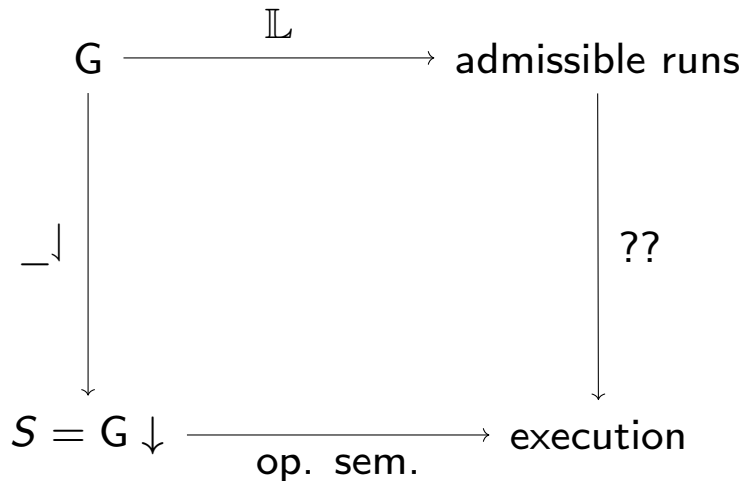
Relating semantics of global views



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Global views as graphs

G-choreographies [25, 16] are regular expressions (on an alphabet of **interactions**) with parallel composition¹

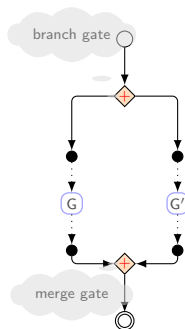
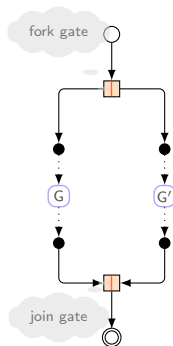
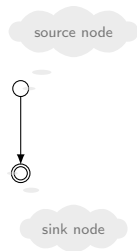
$$G, G' ::= \odot \mid A \rightarrow B : m \mid G \mid G' \mid G ; G' \mid G + G'$$

¹For simplicity, here we omit iteration

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A pomsets semantics

A pomset on a set $\mathcal{E}$ of events is (an isomorphism class of) a labelled partially-order $(\mathcal{E}, \lambda, \leq)$, with

- ▶ $\lambda : \mathcal{E} \rightarrow \mathcal{L}$ a labelling function
- ▶ $\leq \subseteq \mathcal{E} \times \mathcal{E}$ reflexive, anti-symmetric, transitive

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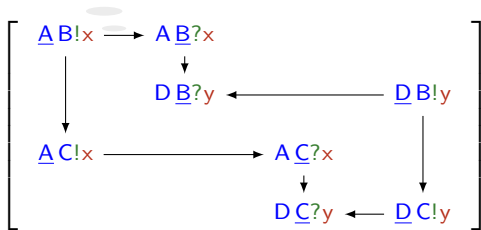
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Our labels : $\mathcal{L} = \mathcal{L}^! \cup \mathcal{L}^?$ where

$$\mathcal{L}^! = \mathcal{C} \times \{!\} \times \mathcal{M} \quad \text{and} \quad \mathcal{L}^? = \mathcal{C} \times \{?\} \times \mathcal{M}$$

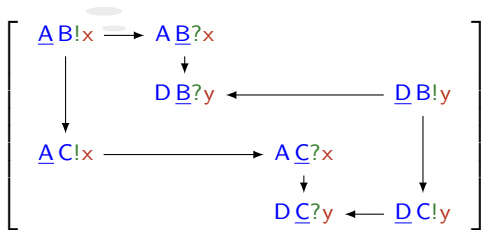
A simple pomset of communication events

meaning $e \leq e'$



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Exercise (An unfair question!)

Find a g-choreography that "corresponds" to the pomset on the left.

From languages of pomsets

The language $\mathbb{L}[r]$ of $r = [\mathcal{E}, \leq, \lambda]$ is the λ -image

if $w = e_1 e_2 e_3$ *and*
 $\lambda(e_i) = A_i B!m$ *then*
 $\lambda(w) = \underline{A_1} B!m \underline{A_2} B!m \underline{A_3} B!m$

From languages of pomsets to languages of choreographies

The language $\mathbb{L}[r]$ of $r = [\mathcal{E}, \leq, \lambda]$ is the λ -image of the (prefixes of) linearisations of r , namely of words $w \in \mathcal{E}^*$ such that for any $0 \leq i \neq j \leq \text{len}(w)$

No duplication $w[i] \neq w[j]$

Causally compliant if $i \leq j$ then $w[j] \not\leq w[i]$

Downward closed for every $e \in \mathcal{E}$, if there is i s.t. $e < w[i]$ then $e \in w$

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The language of $G \in \mathcal{G}$ is

$$\mathbb{L}[G] = \begin{cases} \bigcup_{r \in \llbracket G \rrbracket} \mathbb{L}[r], & \text{if } \llbracket G \rrbracket \text{ is defined} \\ \text{undef}, & \text{otherwise} \end{cases}$$

Pomsets semantics of g-choreographies

We define $\llbracket \cdot \rrbracket : G \mapsto \text{set of pomsets}$ as

$$\llbracket \odot \rrbracket = \{\epsilon\}$$

$$\llbracket A \rightarrow B : m \rrbracket = \left\{ \left[\begin{array}{c} \textcolor{blue}{A} \textcolor{blue}{B} ! \textcolor{red}{m} \rightarrow \textcolor{blue}{A} \textcolor{blue}{B} ? \textcolor{red}{m} \end{array} \right] \right\}$$

$$\llbracket G \mid G' \rrbracket = \{ [\text{disjoint union of } r \text{ and } r'] \mid (r, r') \in \llbracket G \rrbracket \times \llbracket G' \rrbracket \}$$

$$\llbracket G ; G' \rrbracket = \begin{cases} \bigcup_{(r, r') \in \llbracket G \rrbracket \times \llbracket G' \rrbracket} \{ \textcolor{teal}{seq}(r, r') \} & \text{if } \textcolor{red}{ws}(G, G') \\ \text{undef} & \text{otherwise} \end{cases}$$

$$\llbracket G + G' \rrbracket = \begin{cases} \llbracket G \rrbracket \cup \llbracket G' \rrbracket & \text{if } \textcolor{red}{wb}(G, G') \\ \text{undef} & \text{otherwise} \end{cases}$$

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*Assuming
single-
threadness*

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Some simple examples

Semantics of g-choreographies

$$G_1 = A \rightarrow B: x \mid A \rightarrow B: y$$

$$\llbracket G_1 \rrbracket = \left\{ \left[\begin{array}{c} \underline{A}B!x \\ \downarrow \\ \underline{A}\underline{B}?x \end{array} \quad \begin{array}{c} \underline{A}B!y \\ \downarrow \\ \underline{A}\underline{B}?y \end{array} \right] \right\}$$

$$G_2 = A \rightarrow B: x; B \rightarrow A: y$$

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$$G_3 = A \rightarrow B: x + A \rightarrow B: y$$

$$\llbracket G_3 \rrbracket = \left\{ \left[\begin{array}{c} \underline{A}B!x \\ \downarrow \\ \underline{A}\underline{B}?x \end{array} \right], \left[\begin{array}{c} \underline{A}B!y \\ \downarrow \\ \underline{A}\underline{B}?y \end{array} \right] \right\}$$

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$$\begin{aligned} \mathbb{L}[G_1] = \{ & \underline{A}B!x \underline{A}B!y \underline{A}B?x \underline{A}B?y, \\ & \underline{A}B!y \underline{A}B!x \underline{A}B?x \underline{A}B?y, \\ & \underline{A}B!x \underline{A}B!y \underline{A}B?y \underline{A}B?x, \\ & \underline{A}B!y \underline{A}B!x \underline{A}B?y \underline{A}B?x, \\ & \underline{A}B!x \underline{A}B?x \underline{A}B!y \underline{A}B?y, \\ & \underline{A}B!y \underline{A}B?y \underline{A}B!x \underline{A}B?x, \\ & + \text{ all the prefixes} \} \end{aligned}$$

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$$\begin{aligned} \mathbb{L}[G_2] = \{ & \underline{A}B!x \underline{A}B?x \underline{B}A!y \underline{B}A?y \\ & \underline{A}B!x \underline{A}B?x \underline{B}A!y \\ & \underline{A}B!x \underline{A}B?x \\ & \underline{A}B!x \\ & \varepsilon \} \end{aligned}$$

$$G_3 = A \rightarrow B: x + A \rightarrow B: y$$

$$\llbracket G_3 \rrbracket = \left\{ \left[\begin{array}{c} \underline{A}B!x \\ \downarrow \\ \underline{A}B?x \end{array} \right], \left[\begin{array}{c} \underline{A}B!y \\ \downarrow \\ \underline{A}B?y \end{array} \right] \right\}$$

Some simple examples

Semantics of g-choreographies

$$G_1 = A \rightarrow B: x \mid A \rightarrow B: y$$

$$\llbracket G_1 \rrbracket = \left\{ \left[\begin{array}{cc} \underline{A}B!x & \underline{A}B!y \\ \downarrow & \downarrow \\ \underline{A}B?x & \underline{A}B?y \end{array} \right] \right\}$$

$$\begin{aligned} \mathbb{L}[G_1] = \{ & \underline{A}B!x \underline{A}B!y \underline{A}B?x \underline{A}B?y, \\ & \underline{A}B!y \underline{A}B!x \underline{A}B?x \underline{A}B?y, \\ & \underline{A}B!x \underline{A}B!y \underline{A}B?y \underline{A}B?x, \\ & \underline{A}B!y \underline{A}B!x \underline{A}B?y \underline{A}B?x, \\ & \underline{A}B!x \underline{A}B?x \underline{A}B!y \underline{A}B?y, \\ & \underline{A}B!y \underline{A}B?y \underline{A}B!x \underline{A}B?x, \\ & + \text{ all the prefixes} \} \end{aligned}$$

$$G_2 = A \rightarrow B: x; B \rightarrow A: y$$

$$\llbracket G_2 \rrbracket = \left\{ \left[\begin{array}{cc} \underline{A}B!x & \underline{B}A!y \\ \downarrow & \downarrow \\ \underline{A}B?x & \underline{B}A?y \end{array} \right] \right\}$$

$$\begin{aligned} \mathbb{L}[G_2] = \{ & \underline{A}B!x \underline{A}B?x \underline{B}A!y \underline{B}A?y \\ & \underline{A}B!x \underline{A}B?x \underline{B}A!y \\ & \underline{A}B!x \underline{A}B?x \\ & \underline{A}B!x \\ & \varepsilon \} \end{aligned}$$

$$G_3 = A \rightarrow B: x + A \rightarrow B: y$$

$$\llbracket G_3 \rrbracket = \left\{ \left[\begin{array}{c} \underline{A}B!x \\ \downarrow \\ \underline{A}B?x \end{array} \right], \left[\begin{array}{c} \underline{A}B!y \\ \downarrow \\ \underline{A}B?y \end{array} \right] \right\}$$

$$\begin{aligned} \mathbb{L}[G_3] = \{ & \underline{A}B!x \underline{A}B?x, \\ & \underline{A}B!y \underline{A}B?y, \\ & \underline{A}B!x, \\ & \underline{A}B!y, \\ & \varepsilon \} \end{aligned}$$

Some simple examples

Semantics of g-choreographies

$$G_1 = A \rightarrow B: x \mid A \rightarrow B: y$$

$$\llbracket G_1 \rrbracket = \left\{ \left[\begin{array}{cc} \underline{A}B!x & \underline{A}B!y \\ \downarrow & \downarrow \\ \underline{A}B?x & \underline{A}B?y \end{array} \right] \right\}$$

$$\begin{aligned} \mathbb{L}[G_1] = \{ & \underline{A}B!x \underline{A}B!y \underline{A}B?x \underline{A}B?y, \\ & \underline{A}B!y \underline{A}B!x \underline{A}B?x \underline{A}B?y, \\ & \underline{A}B!x \underline{A}B!y \underline{A}B?y \underline{A}B?x, \\ & \underline{A}B!y \underline{A}B!x \underline{A}B?y \underline{A}B?x, \\ & \underline{A}B!x \underline{A}B?x \underline{A}B!y \underline{A}B?y, \\ & \underline{A}B!y \underline{A}B?y \underline{A}B!x \underline{A}B?x, \\ & + \text{ all the prefixes} \} \end{aligned}$$

$$G_2 = A \rightarrow B: x; B \rightarrow A: y$$

$$\llbracket G_2 \rrbracket = \left\{ \left[\begin{array}{cc} \underline{A}B!x & \underline{B}A!y \\ \downarrow & \downarrow \\ \underline{A}B?x & \underline{B}A?y \end{array} \right] \right\}$$

$$\begin{aligned} \mathbb{L}[G_2] = \{ & \underline{A}B!x \underline{A}B?x \underline{B}A!y \underline{B}A?y \\ & \underline{A}B!x \underline{A}B?x \underline{B}A!y \\ & \underline{A}B!x \underline{A}B?x \\ & \underline{A}B!x \\ & \varepsilon \} \end{aligned}$$

$$G_3 = A \rightarrow B: x + A \rightarrow B: y$$

$$\llbracket G_3 \rrbracket = \left\{ \left[\begin{array}{c} \underline{A}B!x \\ \downarrow \\ \underline{A}B?x \end{array} \right], \left[\begin{array}{c} \underline{A}B!y \\ \downarrow \\ \underline{A}B?y \end{array} \right] \right\}$$

$$\begin{aligned} \mathbb{L}[G_3] = \{ & \underline{A}B!x \underline{A}B?x, \\ & \underline{A}B!y \underline{A}B?y, \\ & \underline{A}B!x, \\ & \underline{A}B!y, \\ & \varepsilon \} \end{aligned}$$

Exercise

Now you can verify your answer to the exercise on slide 24.

Some bureaucracy before moving on

$X \uplus Y \triangleq X \times \mathbf{1} \cup Y \times \mathbf{2}$ where

- ▶ $\mathbf{n} \triangleq \{n\}$ for any natural number n
- ▶ $f \otimes \mathbf{n} \triangleq \{(x, n) \mapsto f(x) \mid x \in X\}$ for any f on X
- ▶ $R \otimes \mathbf{n} \triangleq \{((x, n), (y, n)) \mid (x, y) \in R\}$ for any relation $R \subseteq X \times Y$

Given a pomset $r = [\mathcal{E}, \leq, \lambda]$, the pomset of A in r is

$$r|_A = [\mathcal{E}_{r,A}, \leq \cap (\mathcal{E}_{r,A} \times \mathcal{E}_{r,A}), \lambda|_{\mathcal{E}_{r,A}}]$$

where $\mathcal{E}_{r,A} = \{e \in \mathcal{E} \mid \lambda(e) \text{ is performed by } A\}$ is the set of events labelled with actions performed by A .

Well-sequencedness, formally

Let $r = [\mathcal{E}, \leq, \lambda]$ and $r' = [\mathcal{E}', \leq', \lambda']$ and define

$$\begin{aligned}\text{seq}(r, r') &= [\mathcal{E} \uplus \mathcal{E}', \leq_{\text{seq}}, (\lambda \otimes \mathbf{1}) \cup (\lambda' \otimes \mathbf{2})] \\ \leq_{\text{seq}} &= \left((\leq \otimes \mathbf{1}) \cup (\leq' \otimes \mathbf{2}) \cup \bigcup_{\mathbf{A} \in \mathcal{P}} ((\mathcal{E}_{r, \mathbf{A}} \times \mathbf{1}) \times (\mathcal{E}_{r', \mathbf{A}} \times \mathbf{2})) \right)^{\star}\end{aligned}$$

Well-sequencedness, formally

Let $r = [\mathcal{E}, \leq, \lambda]$ and $r' = [\mathcal{E}', \leq', \lambda']$ and define

$$\begin{aligned} \text{seq}(r, r') &= [\mathcal{E} \uplus \mathcal{E}', \leq_{\text{seq}}, (\lambda \otimes \mathbf{1}) \cup (\lambda' \otimes \mathbf{2})] \\ \leq_{\text{seq}} &= \left((\leq \otimes \mathbf{1}) \cup (\leq' \otimes \mathbf{2}) \cup \bigcup_{A \in \mathcal{P}} ((\mathcal{E}_{r,A} \times \mathbf{1}) \times (\mathcal{E}_{r',A} \times \mathbf{2})) \right)^* \end{aligned}$$

Two g-choreographies $G, G' \in \mathcal{G}$ are well-sequenced if

- ▶ $\llbracket G \rrbracket \neq \text{undef}$, $\llbracket G' \rrbracket \neq \text{undef}$, and
- ▶ $ws(r, r')$ for all $r \in \llbracket G \rrbracket$, $r' \in \llbracket G' \rrbracket$

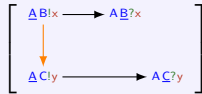
where $ws(r, r')$ if

- ▶ if there are $e \in \mathcal{E}$ and $e' \in \mathcal{E}'$ s.t. $\lambda(e) = A \underline{B} ? m$ and $\lambda'(e') = \underline{A} B ! m$ then $e \leq_{\text{seq}} e'$
- ▶ and $\leq_{\text{seq}(r, r')} \supseteq (\{e \in \mathcal{E} \mid \lambda(e) \in \mathcal{L}^!\} \times \mathbf{1}) \times (\{e \in \mathcal{E}' \mid \lambda'(e) \in \mathcal{L}^?\} \times \mathbf{2})$

Well-sequencedness at work

$A \rightarrow B : x; A \rightarrow C : y$

The "right-input" follows the "left-output"

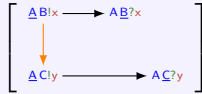


No "right-output"

Well-sequencedness at work

$A \rightarrow B : x; A \rightarrow C : y$

The "right-input" follows the "left-output"



No "right-output"

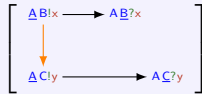
$$r = \left[\text{subj } A \quad \text{subj } B \quad \text{subj } C \quad \text{subj } D \right]$$

$$r' = \left[\text{subj } A \quad \text{subj } B \quad \text{subj } C \right]$$

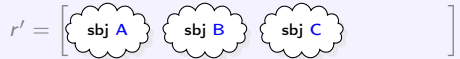
Well-sequencedness at work

$A \rightarrow B : x; A \rightarrow C : y$

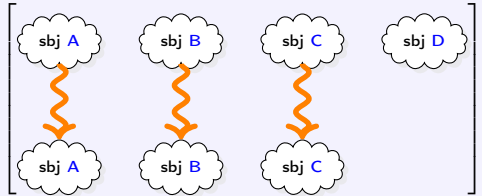
The "right-input" follows the "left-output"



No "right-output"



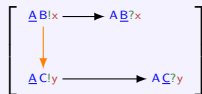
$ws(r, r')$



Well-sequencedness at work

$A \rightarrow B: x; A \rightarrow C: y$

The "right-input" follows the "left-output"

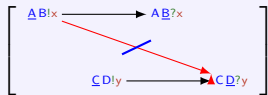


No "right-output"



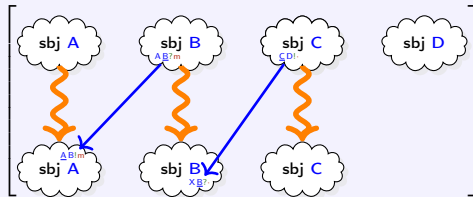
$A \rightarrow B: x; C \rightarrow D: y$

The "right-output" doesn't interfere with "left-inputs"



the "right-input" is not a successor of the "left-output"

$ws(r, r')$



Well-branchedness: an intuition

In a choice $G_1 + G_2$

Well-branchedness: an intuition

In a choice $G_1 + G_2$

- ▶ there should be **at most one active** participant

non-standard

Well-branchedness: an intuition

In a choice $G_1 + G_2$

► there should be **at most one active** participant

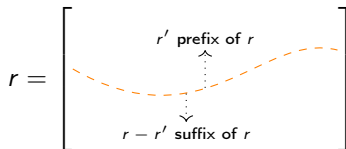
non-standard

► any non-active participant should be **passive**

standard

Problem: Participants do not necessarily "enter" a choice "immediately"

An idea: find a "common part" of the branches for which participants behave uniformly in G_1 and G_2



Some examples

Consider

$$\triangleright A \rightarrow B: \text{int} + A \rightarrow B: \text{str}$$

$$\triangleright A \rightarrow B: \text{int} + \odot$$

$$\triangleright A \rightarrow B: \text{int} + A \rightarrow C: \text{str}$$

$$\triangleright \left(\begin{array}{l} A \rightarrow C: \text{int}; A \rightarrow B: \text{bool} \\ + \\ A \rightarrow C: \text{str}; A \rightarrow C: \text{bool}; A \rightarrow B: \text{bool} \end{array} \right)$$

$$\triangleright G_{\text{😬}} = G_1 + G_2$$

$$G_1 = A \rightarrow C: h; A \rightarrow B: l + A \rightarrow C: y; A \rightarrow B: l; A \rightarrow C: z_1$$

$$\text{where } G_2 = A \rightarrow B: r; A \rightarrow C: h + G_3; A \rightarrow B: z_1 + G_3; A \rightarrow B: z_3$$

$$G_3 = A \rightarrow B: r; A \rightarrow C: y$$

Some examples

Consider

► $A \rightarrow B: \text{int} + A \rightarrow B: \text{str}$

► $A \rightarrow B: \text{int} + \odot$

► $A \rightarrow B: \text{int} + A \rightarrow C: \text{str}$

►
$$\left(\begin{array}{l} A \rightarrow C: \text{int}; A \rightarrow B: \text{bool} \\ + \\ A \rightarrow C: \text{str}; A \rightarrow C: \text{bool}; A \rightarrow B: \text{bool} \end{array} \right)$$

► $G_{\text{😬}} = G_1 + G_2$

$$G_1 = A \rightarrow C: h; A \rightarrow B: l + A \rightarrow C: y; A \rightarrow B: l; A \rightarrow C: z_1$$

where $G_2 = A \rightarrow B: r; A \rightarrow C: h + G_3; A \rightarrow B: z_1 + G_3; A \rightarrow B: z_3$

$$G_3 = A \rightarrow B: r; A \rightarrow C: y$$



Some examples

Consider

► $A \rightarrow B: \text{int} + A \rightarrow B: \text{str}$

► $A \rightarrow B: \text{int} + \odot$

► $A \rightarrow B: \text{int} + A \rightarrow C: \text{str}$

►
$$\left(\begin{array}{l} A \rightarrow C: \text{int}; A \rightarrow B: \text{bool} \\ + \\ A \rightarrow C: \text{str}; A \rightarrow C: \text{bool}; A \rightarrow B: \text{bool} \end{array} \right)$$

► $G_{\text{😊}} = G_1 + G_2$

$$G_1 = A \rightarrow C: h; A \rightarrow B: l + A \rightarrow C: y; A \rightarrow B: l; A \rightarrow C: z_1$$

where $G_2 = A \rightarrow B: r; A \rightarrow C: h + G_3; A \rightarrow B: z_1 + G_3; A \rightarrow B: z_3$

$$G_3 = A \rightarrow B: r; A \rightarrow C: y$$



Some examples

Consider

► $A \rightarrow B: \text{int} + A \rightarrow B: \text{str}$

► $A \rightarrow B: \text{int} + \odot$

► $A \rightarrow B: \text{int} + A \rightarrow C: \text{str}$

►
$$\left(\begin{array}{l} A \rightarrow C: \text{int}; A \rightarrow B: \text{bool} \\ + \\ A \rightarrow C: \text{str}; A \rightarrow C: \text{bool}; A \rightarrow B: \text{bool} \end{array} \right)$$

► $G_{\text{😊}} = G_1 + G_2$

$$G_1 = A \rightarrow C: h; A \rightarrow B: l + A \rightarrow C: y; A \rightarrow B: l; A \rightarrow C: z_1$$

where $G_2 = A \rightarrow B: r; A \rightarrow C: h + G_3; A \rightarrow B: z_1 + G_3; A \rightarrow B: z_3$

$$G_3 = A \rightarrow B: r; A \rightarrow C: y$$



Some examples

Consider

► $A \rightarrow B: \text{int} + A \rightarrow B: \text{str}$

► $A \rightarrow B: \text{int} + \odot$

► $A \rightarrow B: \text{int} + A \rightarrow C: \text{str}$

► $\left(\begin{array}{l} A \rightarrow C: \text{int}; A \rightarrow B: \text{bool} \\ + \\ A \rightarrow C: \text{str}; A \rightarrow C: \text{bool}; A \rightarrow B: \text{bool} \end{array} \right)$

► $G_{\text{😬}} = G_1 + G_2$

where $G_1 = A \rightarrow C: h; A \rightarrow B: l + A \rightarrow C: y; A \rightarrow B: l; A \rightarrow C: z_1$
 $G_2 = A \rightarrow B: r; A \rightarrow C: h + G_3; A \rightarrow B: z_1 + G_3; A \rightarrow B: z_3$
 $G_3 = A \rightarrow B: r; A \rightarrow C: y$



later

Prefix maps & divergence points

The pair of functions (ϕ, ψ) is an A-prefix map of G_1 and G_2 if

$$\begin{array}{ccc} \text{Part}(\llbracket G_1 \rrbracket|_{\mathbf{A}}) & \xrightarrow{\psi} & \mathcal{P}(\text{Pref}(\llbracket G_1 \rrbracket)) \cap \mathcal{P}(\text{Pref}(\llbracket G_2 \rrbracket)) \\ \text{bijection } \phi \downarrow & & \\ & & \text{Part}(\llbracket G_2 \rrbracket|_{\mathbf{A}}) \end{array}$$

Prefix maps & divergence points

The pair of functions (ϕ, ψ) is an A-prefix map of G_1 and G_2 if

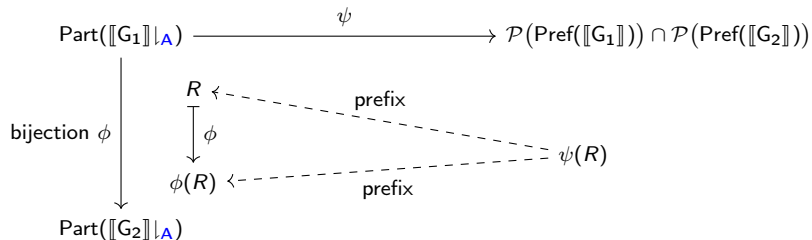
$$\begin{array}{ccc} \text{Part}(\llbracket G_1 \rrbracket|_{\mathbf{A}}) & \xrightarrow{\psi} & \mathcal{P}(\text{Pref}(\llbracket G_1 \rrbracket)) \cap \mathcal{P}(\text{Pref}(\llbracket G_2 \rrbracket)) \\ \downarrow \text{bijection } \phi & & \\ \text{Part}(\llbracket G_2 \rrbracket|_{\mathbf{A}}) & & \end{array}$$

R
 $\downarrow \phi$
 $\phi(R)$

$\psi(R)$

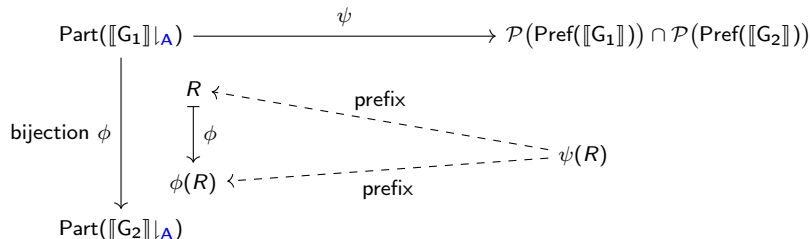
Prefix maps & divergence points

The pair of functions (ϕ, ψ) is an A-prefix map of G_1 and G_2 if



Prefix maps & divergence points

The pair of functions (ϕ, ψ) is an A-prefix map of G_1 and G_2 if



The divergence point of G_1 and G_2 wrt the A-prefix map (ϕ, ψ) is

$$\text{div}_{\phi, \psi}(G_1, G_2) = (L_1, L_2) \quad \text{where} \quad \begin{cases} L_1 = \bigcup_{R \in \text{dom } \phi} \bigcup_{r \in R} \lambda_r(\min(r - \psi(R))) \\ L_2 = \bigcup_{R \in \text{cod } \phi} \bigcup_{r \in R} \lambda_r(\min(r - \psi(\phi^{-1}(R)))) \end{cases}$$

Active & passive components

A is active in $G_1 + G_2$ if there exists an A-prefix map (ϕ, ψ) of G_1 and G_2 with divergence point $\text{div}_{\phi, \psi}(G_1, G_2) = (L_1, L_2)$ s.t.

$$L_1 \cup L_2 \subseteq \mathcal{L}^! \quad L_1 \neq \emptyset \text{ and } L_2 \neq \emptyset \quad L_1 \cap L_2 = \emptyset$$

Active & passive components

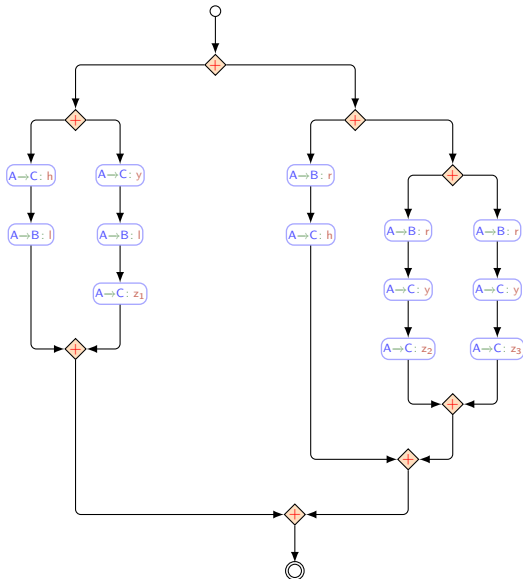
A is active in $G_1 + G_2$ if there exists an A -prefix map (ϕ, ψ) of G_1 and G_2 with divergence point $\text{div}_{\phi, \psi}(G_1, G_2) = (L_1, L_2)$ s.t.

$$L_1 \cup L_2 \subseteq \mathcal{L}^! \quad L_1 \neq \emptyset \text{ and } L_2 \neq \emptyset \quad L_1 \cap L_2 = \emptyset$$

A is passive in $G_1 + G_2$ if there exists an A -prefix map (ϕ, ψ) of G_1 and G_2 with divergence point $\text{div}_{\phi, \psi}(G_1, G_2) = (L_1, L_2)$ s.t.

$$\begin{aligned} L_1 \cup L_2 &\subseteq \mathcal{L}^? & L_1 = \emptyset &\iff L_2 = \emptyset \\ \lambda_r(\mathcal{E}_{r-\psi(R)}) \cap L_2 &= \emptyset \text{ for all } R \in \text{dom } \phi \text{ and } r \in R \\ \lambda_r(\mathcal{E}_{r-\psi(\phi^{-1}(R))}) \cap L_1 &= \emptyset \text{ for all } R \in \text{cod } \phi \text{ and } r \in R \end{aligned}$$

: a difficult choice



$G_{\text{smiley}} = G_1 + G_2$ where

$G_1 = A \rightarrow C: h; A \rightarrow B: l + A \rightarrow C: y; A \rightarrow B: l; A \rightarrow C: z_1$

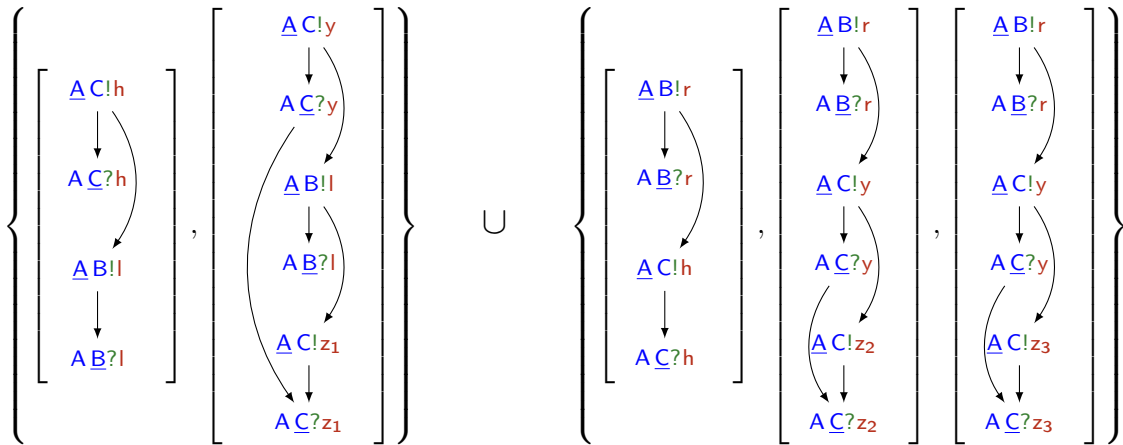
$G_2 = A \rightarrow B: r; A \rightarrow C: h + G_3; A \rightarrow B: z_1 + G_3; A \rightarrow B: z_3$

$G_3 = A \rightarrow B: r; A \rightarrow C: y$

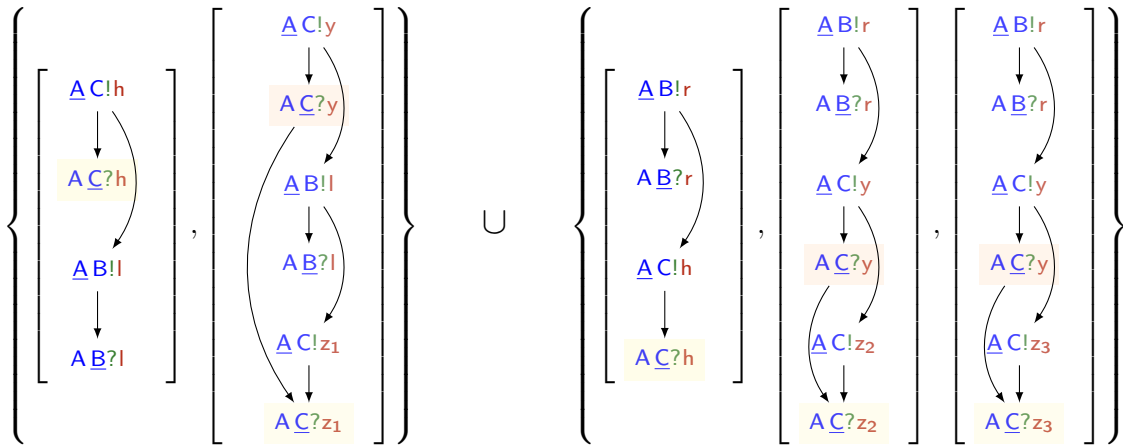
A chooses ... and

- ▶ Whatever B gets, he won't know if A and C exchanged or not **h**
- ▶ Whatever C gets, he won't know if A and B exchanged **l** or **r**

The pomsets of G 😊



The pomsets of G...from C's point of view



Thm If $G \in \mathcal{G}$ with $\llbracket G \rrbracket \neq \perp$ and $S = (\Delta(G \downarrow_A))_{A \in \mathcal{P}}$ then

- ▶ S is deadlock-free
- ▶ $\mathbb{L}[S] \subseteq \mathbb{L}[G]$

The semantics are not equivalent

Projected communicating system can have less behaviours than their protocol due to the FIFO policy on buffers.

Example

Take the g-choreography

$$G = A \rightarrow B : x \mid A \rightarrow B : y$$

we have

$$\underline{A} B ! x \underline{A} B ! y \underline{A} B ? y \underline{A} B ? x \in \mathbb{L}[G] \setminus \mathbb{L}[(\Delta(G \downarrow_A))_{A \in \mathcal{P}}]$$

Thank you!

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